

F728 — The quark module opens, cast as the standing directive says: the whole quark sector is the SVD of ONE overlap matrix on the complexified color off-diagonal $V_{12} \otimes \mathbb{C} = \mathbb{C}^3$ (the SU(3) triplet, Cal K989/29g). Corpus-reconnected (K768), the structure is already there: the Yukawa is an overlap matrix $Y = \langle q_L | \Phi | q_R \rangle$ on $H^2(D_{IV}^5)$; a single condensate makes Φ rank-1, so Y is rank-1 at leading order — the top is the one leading singular value, $y_t = \|P_L O\| \cdot \|P_R O\| \leq 1$ (Cauchy–Schwarz, DERIVED ceiling), and the ENTIRE hierarchy (c, u, b, s, d) + the CKM mixing are the off-rank-1 corrections (Σ – rank-1), Tier-2 by construction. The count is Derived (§132 strata transfer — domain-level, K989/K990). The honest tiers are pre-registered UP FRONT, and the discipline that carries the module is stated first: on the colored/continuous sector, a quark matching a clean lepton-style closed form is a RED FLAG, not a win — the honesty is the credibility. The single decisive computation is unchanged (K768): pin O ’s direction non-circularly from the boundary (F603: O = the (2,2) bi-doublet, spherical), then $\langle t | O \rangle$ decides $y_t = 1$ (parallel, Derived) vs < 1 (a computed CG). This note is the module OPENER + explicit-matrix setup + tier bar; Elie builds the $\mathbb{C}^3$ SVD numeric.

Lyra, Wed 2026-07-29. The quarks open on the real structure (the colored corner of the one overlap matrix), not the borrowed lepton method. Standing directive first: recast to explicit matrices on D_{IV}^5 — here the matrix is a 3×3 complex SVD on $V_{12} \otimes \mathbb{C}$. Corpus-first (K768 rank-1 + Tier-2; Cal’s $\mathbb{C}^3$ home; F603 O -direction; F718 address), tiers pre-registered before any number.

The home (Cal K989/29g — the correct linear-algebra address)

Quark fermions live in $V_{12} \otimes \mathbb{C} = \mathbb{C}^3$: V_{12} is the Peirce color off-diagonal ($\dim = N_c = 3$, dimensionally forced), and SU(3) genuinely acts on its complexification $\mathbb{C}^3$ (the fundamental triplet), grounded in the domain’s Hermitian structure. This is why the quarks are the *messy* corner: colored modes live off the color-neutral frame, so their mass operator does not commute with the idempotent frame → no clean eigenvalue split (unlike the colorless leptons, F727) → an SVD, not a diagonalization.

The object (K768, corpus — the SVD is already there)

The Yukawa/mass operator is the overlap matrix on the quark modes:

$$Y_{ij} = \langle q_L^i | \Phi | q_R^j \rangle \quad \text{on } H^2(D_{IV}^5), \quad \Phi = \text{condensate (rank-1 from one } O \text{)}.$$

- Leading (rank-1): $Y = a \otimes b$ exactly ONE non-zero singular value only the top is massive at leading order, $y_t = \|P_L O\| \cdot \|P_R O\| \leq 1$ (Cauchy–Schwarz). Ceiling DERIVED. - Subleading (Φ – rank-1): charm, up, bottom, strange, down, and the CKM mixing are the off-rank-1 singular values + vectors — subleading deviations of Φ from rank-1. Tier-2 by construction — and this *explains* why the up-ladder and mixing angles back-solved / died on ~8 routes: they are corrections, not leading identities.

The SVD picture, explicit: $Y = U \Sigma V^\dagger$ on $\mathbb{C}^3$ — Σ = singular values (masses), U/V = the mixing rotations (CKM). Top = Σ_1 (rank-1 leading); the rest = the correction spectrum.

Pre-registered honest tiers (the bar Keeper audits against — stated BEFORE the numbers)

quark-sector claim	expected tier	basis
generation count = 3	Derived	§132 strata (domain-level, transfers; K990)
top-quark ceiling $y_t \leq 1$	Derived	Cauchy–Schwarz on the rank-1 overlap (K768)
$y_t = 1$ (exact)	Supported $\rightarrow$ Derived if $O \nsubseteq \text{top}$	the decisive O-computation (K768)
$m_s/m_d \approx 20$	Tier-1 (specific ratio)	the pre-registered ratio bar
individual masses (c, u, b, s, d)	Tier-2 structural	off-rank-1 corrections; soft/running/continuous
CKM mixing angles	Tier-2 structural	off-rank-1 corrections of the same O (K704)
any clean closed form for a single colored quark mass	RED FLAG	colored sector is continuous — a lepton-style clean formula signals over-fit, not derivation

The discipline, first-class: the credibility of the quark module IS the honesty of these tiers. We do not chase clean formulas for the light quarks; we expect a structured Tier-2 correction spectrum off one condensate, and we reject over-fits loudly. A colored quark landing on a $(24/\pi^2)$ -style clean number is a coincidence to be exposed, not a win to be banked.

The single decisive computation (unchanged, K768 — the one dig, four payoffs, F718)

Pin O's direction non-circularly from the conformal boundary (F603: $O = \text{the } SO(5)\text{-vector } (2,2) \text{ bi-doublet, spherical, boundary-reaching — pinned from quantum numbers, target-innocent}$), then compute the overlaps: - $\langle t|O \rangle \rightarrow y_t = 1$ (top $\nsubseteq O$, Derived) or a computed $CG < 1$ (0.992 reads as near-but-not-exact — the 0.8% gap is informative). - $\langle c|O \rangle, \langle u|O \rangle$ and the down-type $\rightarrow$ the leading hierarchy (the off-rank-1 structure). - The same O re-derives the mixing (CKM = the off-rank-1 correction angles, K704). - F718 address dig: the quark-generation radial/boundary-distance controls both the ordering (Derived, F603/K768) and the precision-loss ladder (the blur = why quarks are Tier-2) — one address computation, four payoffs.

Handoffs

- @Elie — build the explicit 3×3 complex SVD on $\mathbb{C}^3 = V_{12} \otimes \mathbb{C}$: (i) confirm rank-1 $\Phi \nsubseteq$ one leading singular value (top), (ii) form the off-rank-1 correction spectrum for the up/down ladders, (iii) read the singular vectors as CKM. Source the Peirce/color structure (T2511, FK Ch VI) — do not reconstruct. Report singular values, not verdicts; the tier bar above is pre-registered.
- @Cal — audit every quark tier against the confined-bulk law + the pre-registered bar: count Derived, top-ceiling Derived, $m_s/m_d=20$ Tier-1, the rest Tier-2, over-fits rejected. Your $\mathbb{C}^3$ home (SU(3) on the complexification) is the setup's foundation — confirm the overlap matrix is genuinely on $V_{12} \otimes \mathbb{C}$, not smuggling the frame.
- @Keeper — rule each quark singular value as it lands against the bar; hold the over-fit red-flag line (a clean colored formula is scrutinized, not banked). The count is already Derived (K990); the top-ceiling is Derived (K768); everything else is the honest Tier-2 spectrum.
- @Grace — pin O's direction from the boundary (F603, non-circular) is the decisive input; the mixing = off-rank-1 corrections (K704) rides the same O.
- @Casey — the quarks open exactly on the structure, not a transplant. The whole quark sector is one matrix's SVD, and it's built to be honest: the top quark is the one clean thing (it's the leading, rank-1 mode, and its ceiling — that its coupling can't exceed 1 — is derived), and everything else (the lighter quark masses, the mixing) is one *correction spectrum* off a single condensate — genuinely Tier-2, soft and running, and we say so. The one computation that closes the clean part is pinning the condensate's direction and asking

whether the top lines up with it (giving $y_t = 1$) — the same by-hand target we've had. And the guardrail is stated first: if a light quark mass suddenly matches a tidy formula like the muon's, that's a red flag to expose, not a trophy — because on the colored sector, being honest about what's Tier-2 is the whole credibility.

Notes only; no toy/theorem claimed (Elie owns the SVD numeric). F728 (quark module opener): quark sector = SVD of ONE overlap matrix $Y = \langle q_L | \Phi | q_R \rangle$ on $V_{12} \otimes \mathbb{C} = \mathbb{C}^3$ (Cal's SU(3)-triplet home). Rank-1 Φ (K768) $\Rightarrow$ top = leading singular value, $y_t \leq 1$ Cauchy-Schwarz DERIVED; hierarchy + CKM = off-rank-1 corrections, Tier-2 by construction. Count Derived (§132). Pre-registered tiers UP FRONT: count Derived, top-ceiling Derived, $m_s/m_d=20$ Tier-1, rest Tier-2, clean colored formula = RED FLAG. Decisive computation: pin O (F603, non-circular) $\rightarrow \langle t|O \rangle \rightarrow y_t=1$ vs <1 ; same O re-derives mixing; F718 address = one dig four payoffs. Elie builds $\mathbb{C}^3$ SVD; Keeper rules vs bar; Cal audits $\mathbb{C}^3$ home + over-fits. Honesty = credibility on the colored sector. — Lyra